

Physics of Climate Change — Fall 2025 — Tentative Course Calendar

Week	Lecture Topic (Tue)	Laboratory (Wed)	Discussion (Thu)
Aug 26–28	Temperatures and CO ₂ over Time	Measurement and Uncertainty	Earth's History of Temp
Sep 2–4	Contemporary Climate Change	Global Average Temperature	Weather Indicators of Clim Ch
Sep 9–11	Energy and Equilibrium	Mechanical equilibrium (leaky buckets I)	Energy and Energy Transf
Sep 16–18	EM Radiation (light) and Blackbody	All the light you cannot see (BB rad)	BBody Rad & Earth Temp
Sep 23–25	Atmosphere and Greenhouse Eff I	Heating by radiation	Earth temp: w/Atmosphere
Sep 30 – 2	Atmosphere and Greenhouse Eff II	Absorption of radiation / Blanketing	Atmosph Absorption Spectra
Oct 7–9	Water and Phase Transitions I	Cloud Formation	Evaporation/Condensation
Oct 14–16	Water and Phase Transitions II	Phase Transitions	Earth Temp: Rad-Conv Equil
Oct 21–23	Fall Break (no class)	No Lab (probably)	Intro to Milankovitch Cycles
Oct 28–30	Paleo Perturbations of Temperature	Milankovich Cycles	More Paleo and Milankovitch
Nov 4–6	Climate Sensitivity	IR absorption by GH gases	CO ₂ absorption line
Nov 11–13	Timescales of Climate Change	Equilibration: buckets and radiation	Rates of Change, Timescales
Nov 18–20	Modeling Climate Change	Computational Climate Models	
Nov 25–27	Oceans and Ice	Thanksgiving (no class)	
Dec 2–4	work on project	Ice and Thermal Exp of H ₂ O	work on project
FINAL EXAM TIMESLOT: Thursday, December 11, 2025, 10am–Noon (Poster Presentations)			